# AQ2.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.
**Level 1:**

*1 point*
Match the matrices in Column A with their inverses in Column B.

- [ ] a → → iii
- [ ] a → → ii
- [ ] b → → iii
- [ ] b → → i
- [ ] c → → i
- [ ] c → → ii

## No, the answer is incorrect.
## Score: 0
## Accepted Answers:
a → → ii
b → → iii
c → → i

**(Use the below information to answer questions 2 and 3).**
Consider a system of linear equations

$$2x_1 - x_2 = 3$$
$$x_1 - x_3 = 3$$
$$x_2 - x_3 = 2$$

2x1 − x2 = 3x1 − x3 = 3x2 − x3 = 2

Let the matrix representation of the above system be $Ax = b$Ax = b, where $A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}$A = [2−1 0 1 0 −1 0 1 −1], $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = [x1x2x3], and $b = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix}$b = [332].

*1 point*
Choose the set of correct options.

- [ ] $A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$A−1 = [211−1−1000−1]

- [ ]

$A^{-1}$ = [1−11 1−22 1−21]A−1 = [111−1−2−2121]

☐

Adjoint of the matrix $A$A is [1−11 1−22 1−21] [111−1−2−2121]

☐

$det(A)$ = 1det(A) = 1.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$A^{-1}$ = [1−11 1−22 1−21]A−1 = [111−1−2−2121]

Adjoint of the matrix $A$A is [1−11 1−22 1−21] [111−1−2−2121]

$det(A)$ = 1det(A) = 1.

*1 point*

Choose the set of correct options.

☐

$x_1$ = − 2x1 = −2.

☐

$x_2$ = 1x2 = 1.

☐

$x_3$ = − 1x3 = −1.

☐ None of the above.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$x_2$ = 1x2 = 1.

$x_3$ = − 1x3 = −1.

*1 point*

Choose the set of correct options.

☐

A system of linear equations $Ax$ = $b$Ax = b is called a homogeneous system of linear equations if $b \neq 0$b = 0.

☐

A system of linear equations $Ax$ = $b$Ax = b is called a non-homogeneous system of linear equations if $b \neq 0$b = 0

☐

If $v$v is a solution of the system of linear equations $Ax$ = $b$Ax = b, then $\frac{1}{c}v$c1v is a solution of system of linear equations $cAx$ = $b$cAx = b, where $c \neq 0$c = 0.

☐

Let $Ax = b$Ax = b be a system of linear equations. If $A$A is invertible, then $adj(A)x = b$adj(A)x = b also has a solution.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
A system of linear equations $Ax = b$Ax = b is called a non-homogeneous system of linear equations if $b \neq 0$b = 0
If $v$v is a solution of the system of linear equations $Ax = b$Ax = b, then $\frac{1}{c}v$c1v is a solution of system of linear equations $cAx = b$cAx = b, where $c \neq 0$c = 0.
Let $Ax = b$Ax = b be a system of linear equations. If $A$A is invertible, then $adj(A)x = b$adj(A)x = b also has a solution.

Suppose $C$C is the cofactor matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$. Let $C_{ij}$ Cij be the $ij$ij-th element of the cofactor matrix $C$C. Answer the questions 5,6 and 7.

Find the value of $C_{12}$ C12.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 1
*1 point*
Find the value of $C_{22}$ C22.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) -3
*1 point*
Find the sum of the elements in the third row of $C$C.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 0
*1 point*

**Level 2:**

*1 point*
Which of the following square matrices of order 3 are the same as their adjoint matrices?
- [ ] Identity matrix.
- [ ] Zero matrix.
- [ ] Any scalar matrix.

☐ Any diagonal matrix.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Identity matrix.

Zero matrix.

*1 point*

Choose the set of correct options.

[Hint: For the first option, find out the adjoint matrix of an arbitrary square upper triangular matrix of order 22, i.e., take the matrix $\begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$[a0bd] and find out its adjoint matrix.]

☐

The adjoint of a $2 \times 2$2 × 2 real upper triangular matrix is an upper triangular matrix.

☐

There exists a square matrix of order 33, such that $A = A^{-1} = adj(A)$A = A−1 = adj(A).

☐

If $A = A^{-1}$A = A−1, then $det(A)$det(A) must be 11.

☐

If $A = A^{-1}$A = A−1, then $A$A must be an identity matrix.

☐

If $A = A^{-1}$A = A−1, then $A^2 = I$A2 = I.

☐

If $A^{-1} = adj(A)$A−1 = adj(A), then $det(A)$det(A) must be 11.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The adjoint of a $2 \times 2$2 × 2 real upper triangular matrix is an upper triangular matrix.

There exists a square matrix of order 33, such that $A = A^{-1} = adj(A)$A = A−1 = adj(A).

If $A = A^{-1}$A = A−1, then $A^2 = I$A2 = I.

If $A^{-1} = adj(A)$A−1 = adj(A), then $det(A)$det(A) must be 11.

*1 point*

If $A$ and $B$A and B are squares matrices of order 22, then choose the set of correct options.

[Hint: Find out the adjoint matrix of an arbitrary square matrix of order 22, i.e., take the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$[ac bd] and find out its adjoint matrix. ]

☐

$adj(AB) = adj(A)adj(B)$adj(AB) = adj(A)adj(B).

☐

$adj(AB) = adj(B)adj(A)$adj(AB) = adj(B)adj(A).

☐

$adj(A + B) = adj(A) + adj(B)$adj(A + B) = adj(A) + adj(B).

☐

$adj(A^T) = adj(A)^T$ adj(AT) = adj(A)T .

☐

$adj(A^{-1}) = adj(A)^{-1}$ adj(A−1) = adj(A)−1.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$adj(AB) = adj(B)adj(A)$adj(AB) = adj(B)adj(A).

$adj(A + B) = adj(A) + adj(B)$adj(A + B) = adj(A) + adj(B).

$adj(A^T) = adj(A)^T$ adj(AT ) = adj(A)T .
$adj(A^{-1}) = adj(A)^{-1}$ adj(A−1) = adj(A)−1.

Check Answers

Your score is: 0/10